NLP for Teens

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NLP pipeline

It’s just like any other pipeline where you put data in, do some steps, and get results out. In this NLP version, you start with text, do a bit of cleaning, break it into pieces, turn words into numbers

Data acquisition

Data acquisition means getting the raw text you want to work with.

Cleanup

Removing unwanted data

NLP pipeline

Data Acquisition: This is just collecting your raw text from wherever it is.

Cleaning: A quick tidy-up—removing unwanted bits.

Tokenization: Breaking your text into smaller pieces, like words.

Vectorization: Turning those words into numbers so the computer can understand them.

Stemming vs Lemmatization

Stemming and lemmatization both aim to reduce words to their base forms, but they do it differently:

Stemming is a bit like a shortcut. It chops off word endings to get to the root form, but it doesn’t always result in a real word. For example, "running" might become "runn.“ (running → runn)

Lemmatization is more precise. It reduces words to their dictionary form (the lemma), so "running" would become "run.“ (running → run)

In short, stemming is faster but less accurate, while lemmatization is more accurate but a bit slower.

Assignment

- Process the sentences with spaCy to print each token and its lemma, then use NLTK to stem the same sentences and compare the results.
- Sentences:
  - The leaves on the trees are falling.
  - She ran quickly to catch the bus.
  - We are learning about natural language processing.

Summary

- ✓ Tokenization: breaking text into words.
- ✓ Lemmatization: converting words to base form.
- ✓ Stemming: cutting words to rough roots.
- ✓ spaCy_pipeline: fast, modern NLP processing.
- ✓ NLTK_stemming: simple rule-based stemmer.

Thank You

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